

Graph Topology-Guided State-Space and
Mixture-of-Experts for Dynamic Brain
Connectome Classification and Spatiotemporal
Pathological Localization in Alzheimer’s Disease

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1 Detailed Derivation of the Fourier Graph Operator

In the main text, we introduced the Fourier Graph Operator (FGO) to extract intra-window asynchronous features in the frequency domain. Here, we provide the complete formulation and derivation to demonstrate its mathematical equivalence to the graph convolution.

For the BOLD signal matrix $X_{win}^{(i)} \in \mathbb{R}^{N \times L}$ within the i -th sliding window ($i = 1, \dots, W$), we construct a fully connected hypervariate graph $\mathcal{G}_{hv}^{(i)} = (X^{\mathcal{G}^{(i)}}, A^{\mathcal{G}^{(i)}})$. We project the node features $X^{\mathcal{G}^{(i)}}$ into a d -dimensional embedding $X^{\mathcal{G}_*^{(i)}} \in \mathbb{R}^{(N * L) \times d}$.

To perform graph convolution operations in the frequency domain, we define a Fourier Graph Operator (FGO). For each window, we introduce an independent learnable weight matrix $W^{(i)} \in \mathbb{R}^{d \times d}$ to derive a tailored Green's kernel $\kappa^{(i)}: [(N * L)] \times [(N * L)] \rightarrow \mathbb{R}^{d \times d}$, where $\kappa^{(i)}[m, n] := A_{mn}^{\mathcal{G}^{(i)}} \circ W^{(i)}$ and $\kappa^{(i)}[m, n] = \kappa^{(i)}[m - n]$. The Fourier Graph Operator (FGO) is defined as follows:

$$\mathcal{S}^{(i)} = \mathcal{F}(\kappa^{(i)}) \in \mathbb{C}^{d \times d} \quad (1)$$

where $\mathcal{F}$ denotes the Discrete Fourier Transform (DFT). According to the Convolution Theorem [1], the Fourier transform of the convolution of two signals equals the pointwise product of their Fourier transforms in the frequency domain. Therefore, the multiplication between $\mathcal{F}(X^{\mathcal{G}_*^{(i)}})$ and the Fourier Graph Operator $\mathcal{S}^{(i)}$ in Fourier

space is expressed as:

$$\begin{aligned}
\mathcal{F}\left(X^{\mathcal{G}^{(i)}}_*\right)\mathcal{S}^{(i)} &= \mathcal{F}\left(X^{\mathcal{G}^{(i)}}_*\right)\mathcal{F}\left(\kappa^{(i)}\right) \\
&= \mathcal{F}\left(\left(X^{\mathcal{G}^{(i)}}_* * \kappa^{(i)}\right)[m]\right) \\
&= \mathcal{F}\left(\sum_{n=1}^{(N*L)} X^{\mathcal{G}^{(i)}}_*[n]\kappa^{(i)}[m-n]\right) \\
&= \mathcal{F}\left(\sum_{n=1}^{(N*L)} X^{\mathcal{G}^{(i)}}_*[n]\kappa^{(i)}[m,n]\right), \quad \forall m \in [(N*L)]
\end{aligned} \tag{2}$$

where $X^{\mathcal{G}^{(i)}}_* * \kappa^{(i)}$ denotes the convolution operation in the spatial domain. Based on the definition $\kappa^{(i)}[m,n] := A^{\mathcal{G}^{(i)}}_{mn} \circ W^{(i)}$, we can derive $\sum_{n=1}^{(N*L)} A^{\mathcal{G}^{(i)}}_{mn} X^{\mathcal{G}^{(i)}}_*[n]W^{(i)} = A^{\mathcal{G}^{(i)}}_* X^{\mathcal{G}^{(i)}}_* W^{(i)}$. Accordingly, we derive the final convolution formula:

$$\mathcal{F}\left(X^{\mathcal{G}^{(i)}}_*\right)\mathcal{S}^{(i)} = \mathcal{F}\left(A^{\mathcal{G}^{(i)}}_* X^{\mathcal{G}^{(i)}}_* W^{(i)}\right) \tag{3}$$

This derivation demonstrates that the recursive pointwise multiplication between the transformed features and the Fourier Graph Operator $\mathcal{S}^{(i)}_{0:k}$ in our FourierGNN⁽ⁱ⁾ module is mathematically equivalent to multi-order graph convolution.

2 Experiment Settings

To evaluate the diagnostic performance of the proposed framework across different stages of Alzheimer’s disease (AD), we conducted extensive comparative experiments. The experiments covered four clinically significant binary classification tasks: AD diagnosis (AD vs. NC), early MCI identification (LMCI vs. NC), differentiation of late-stage disease progression (LMCI vs. AD), and MCI subtype differentiation (EMCI vs. LMCI).

To capture the temporal dynamics, the rs-fMRI time series were segmented using a sliding window technique with a window size of 10 and a stride of 5. For the construction of functional brain networks and patch networks, the adjacency matrix threshold was uniformly set to the 80th percentile. Regarding the network architecture, the number of FourierGNN layers was uniformly set to 1. During training, we employed the Adam optimizer with an initial learning rate of 0.0001, a batch size of 16, and a maximum of 100 epochs. To ensure robustness, we performed five-fold cross-validation on the training data. For each fold, the model parameters yielding the highest accuracy (ACC) on the validation set were saved and subsequently evaluated on the independent test set. We report three metrics to evaluate performance: Accuracy (ACC), F1-score (F1), and Area Under the ROC Curve (AUC).

To comprehensively evaluate the efficacy of our proposed framework, we selected several state-of-the-art methods for comparison. To ensure a fair evaluation, all baseline methods were implemented using the optimal hyperparameter configurations recommended in their respective original publications. The selected methods are categorized as follows: (1) Static deep learning methods, which include spatial-focused Convolutional Neural Networks (CNN), Graph Convolutional Networks (GCN) [2], and advanced GNN-based architectures such as BrainGNN [3], HFBN [4], and ACIFBN [5]; (2) Dynamic deep learning methods, which are designed to capture rs-fMRI time-varying characteristics using sliding windows or attention mechanisms, including MSTGAC [6] and contemporary Transformer-based architectures (Transformer [7], BNT [8], Com-BNT [9], and BolT [10]); and (3) Time-Frequency Fusion Baseline, specifically FTF-GNN [11], a previously proposed static GNN model that integrates temporal and spectral features.

- **BrainGNN [3]:** The network was constructed with 2 Graph Neural Network layers and an initial feature embedding dimension of 200. The node pooling ratio was set

to 0.5. Crucially, to guide the ROI-selection process, the regularization weights for both pooling entropy (λ_3, λ_4) and consistency (λ_5) were uniformly set to 0.1.

- **HFBN** [4]: The hierarchical pooling mechanism was configured with a single pooling layer, node assignment ratios of 0.35, and a network channel multiplier of 8 to extract topological properties.
- **ACIFBN** [5]: The core architecture was constructed using a 5-layer spatiotemporal configuration (feature channels transitioning from 6 to 12). To capture long-range temporal dependencies, early layers employed exceptionally large chunk kernel sizes (32 and 64), while subsequent layers utilized a convolution stride of 2 for temporal downsampling. The base temporal convolution kernel size was set to 3, and the graph sparsity control parameter (α) was fixed at 2.0.
- **MSTGAC** [6]: The spatio-temporal contrastive network processed the dynamic functional connectivity using 17 temporal windows. To generate positive views for contrastive learning, topological dropout probabilities were set to 0.075 and 0.125, respectively. Spatial features were aggregated via an attention mechanism configured with an input dimension of 128 and a hidden dimension of 16. Subsequently, temporal evolution was captured using a 2-layer LSTM, and the entire network was optimized via the InfoNCE loss with a temperature coefficient (τ) of 0.4.
- **BNT** [8]: The architecture consisted of 4 Transformer encoding layers with a hidden embedding dimension of 128. An identity positional encoding strategy with an embedding dimension of 90 was utilized. The node clustering sizes across the hierarchical stages were set to 116 and 100, governed by an orthogonal projection assignment mechanism.
- **Com-BNTF** [9]: The model was constructed with 4 Transformer layers and 1 Multi-Head Self-Attention (MHSA) block utilizing 4 attention heads. The hidden embedding dimension was set to 128, with a spatial embedding dimension of 116. The community-aware hierarchical clustering sizes were defined as 116 and 8, which

were also optimized via an orthogonal projection assignment mechanism to ensure robust cluster representations.

- **Com-BNTF** [9]: The model was constructed with 4 Transformer layers and 1 Multi-Head Self-Attention (MHSA) block utilizing 4 attention heads. The hidden embedding dimension was set to 128, with a spatial embedding dimension of 116. The community-aware hierarchical clustering sizes were defined as 116 and 8, which were also optimized via an orthogonal projection assignment mechanism to ensure robust cluster representations.
- **BolT** [10]: The Fused Window Transformer was configured with 4 layers and 36 attention heads (with a head dimension of 20). To capture dynamic temporal transitions, the focal window size was set to 20, utilizing a shift coefficient of 0.4 and a fringe coefficient of 2, governed by an "expand" focal rule. Furthermore, a dropout rate of 0.1 and an MLP ratio of 1.0 were applied to prevent overfitting.
- **FTF-GNN** [11]: As a time-frequency fusion baseline, the architecture was configured with task-specific hyperparameters to optimize performance. Specifically, the number of FourierGNN layers was set to 1 for the AD vs. NC task, and uniformly set to 5 for the remaining three classification tasks. Furthermore, the adjacency matrix thresholds governing graph construction were precisely calibrated to 0.55, 0.5, 0.6, and 0.6 across the four respective tasks.

3 Parameter Sensitivity Analysis

The total number of experts E and the sparse selection number Top_K in the Graph Topology-guided Mixture of Experts (GT-MoE) module are critical to model performance. To avoid the prohibitive computational cost of an exhaustive grid search, we adopted a stepwise search strategy to determine the optimal parameter configuration.

First, with Top_K fixed at 2, we evaluated the impact of varying E within the range $\{2, 3, 4, 5, 6\}$ on classification performance. As illustrated in Fig. 1 (Left), the model's

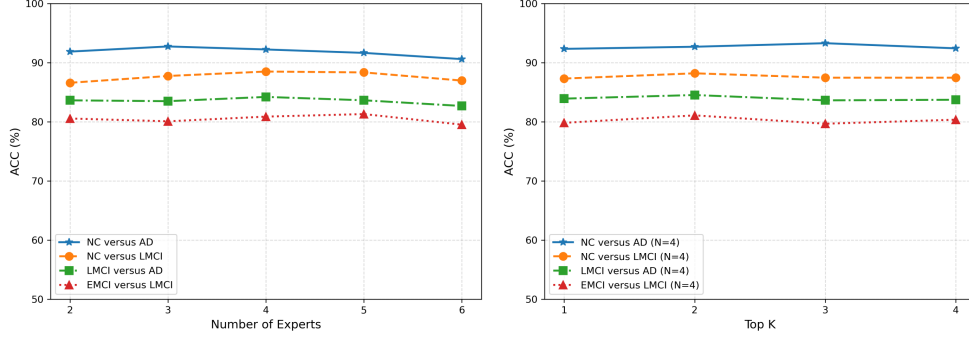
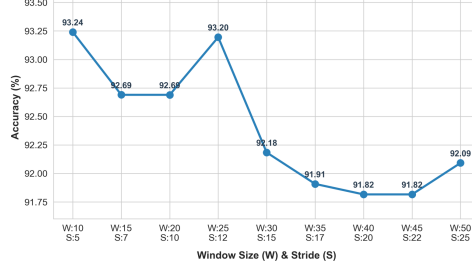
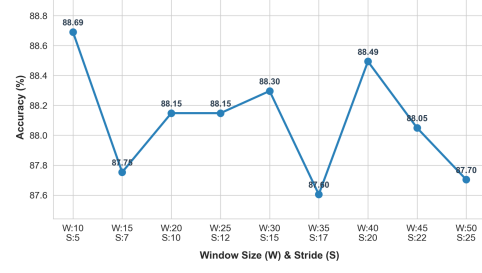


Fig. 1: Parameter sensitivity analysis on total number of experts E and the sparse selection number Top_K

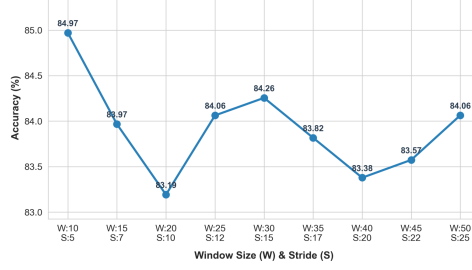
accuracy across the four classification tasks exhibits a trend of initially increasing and then decreasing as the number of experts rises. When $E = 4$, the model achieves optimal or near-optimal performance on most tasks (particularly in the NC vs. AD and LMCI vs. AD task). When the number of experts is too small (e.g., $E = 2$), model capacity is insufficient to capture the complex heterogeneous patterns of brain networks. Conversely, when the number of experts is excessive (e.g., $E = 6$), performance degrades, likely due to overfitting from parameter redundancy or an overly dispersed routing mechanism that leaves individual experts with insufficient training data. Subsequently, with E fixed at 4, we investigated the influence of Top_K within the range $\{1, 2, 3, 4\}$. Fig. 1 (Right) indicates that the model is most robust when $K = 2$. When $K = 1$, the model degenerates into hard routing selection, potentially losing complementary information from sub-optimal experts. When the Top_K value is too large (e.g., $K = 4$), the benefits of sparse routing are mitigated, increasing parameter overhead and introducing noise from irrelevant experts. Balancing computational efficiency with diagnostic accuracy, we set $E = 4$ and $K = 2$. Following the determination of the MoE configuration, we evaluated the impact of sliding window parameters—specifically, window size (W) and stride (S)—on capturing dynamic temporal evolution. As illustrated in Fig. 2, smaller window and stride configurations



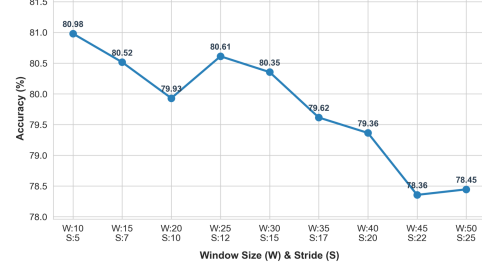
(a) AD vs. NC



(b) LMCI vs. NC



(c) LMCI vs. AD



(d) EMCI vs. LMCI

Fig. 2: Parameter sensitivity analysis on window size W and stride S

generally yield superior performance. The model consistently achieves the highest classification accuracy across all four tasks when $W = 10$ and $S = 5$. This indicates that compact window and stride settings preserve fine-grained spatiotemporal details, enabling the model to precisely capture highly transient yet discriminative pathological micro-states within the brain network. Conversely, as the window size increases, these subtle transient connectivity features are gradually smoothed out by long-term averaging effects, leading to a moderate decline in classification performance. Nevertheless, it is noteworthy that the performance degradation across larger window and stride combinations remains marginal. For instance, in the most challenging EMCI vs. LMCI task, the maximum accuracy drop is less than 3%; similarly, in the AD vs. NC task, the accuracy remains stable above 91.8% even under extreme configurations (e.g., $W = 40$ or 50). This minimal fluctuation demonstrates that the proposed architecture is highly robust to time-series partitioning parameters, deriving its diagnostic

efficacy from its intrinsic spatiotemporal feature extraction mechanism rather than relying on a strictly tuned data split. Consequently, to maximize the capture of fine-grained dynamic brain network features, we uniformly set $W = 10$ and $S = 5$ across all tasks.

4 Time Complexity Analysis

To theoretically demonstrate the computational efficiency of the model, we analyze the time complexity of its core modules. Assume that the input rs-fMRI sequence is segmented into W sliding windows, where each window contains N brain region nodes (ROIs) with a sequence length L and a feature embedding dimension d . First, in the phase of time-domain feature extraction, the time complexity of calculating the Pearson Correlation Coefficient (PCC) between ROIs within each window is $\mathcal{O}(W \cdot N^2 \cdot L)$. Simultaneously, the time-domain Mamba module models long-range temporal dependencies of the dynamic functional features across windows, which similarly incurs a complexity of $\mathcal{O}(W \cdot N \cdot L \cdot d)$. In the frequency-domain feature extraction module, the FourierGNN, which encompasses the Discrete Fourier Transform (DFT), its inverse (IDFT), and frequency-domain matrix multiplication, carries a complexity of $\mathcal{O}(W \cdot (Nd \log N + Nd^2))$. Subsequently, the frequency-domain representations are mapped back to the time domain and fed into the frequency-domain Mamba module, requiring a time complexity of $\mathcal{O}(W \cdot N \cdot L \cdot d)$. Finally, in the Graph Topology-guided Mixture of Experts (GT-MoE) module, the gated routing network calculates probability scores to assign experts for each node with a complexity of $\mathcal{O}(W \cdot N \cdot d \cdot E)$ (where E is the total number of experts). Benefiting from the Top- K sparse routing mechanism, only a small number of expert modules are activated. The computational complexity for the activated GCN experts to aggregate graph topological features is strictly controlled at $\mathcal{O}(W \cdot N^2 \cdot d)$.

In summary, the total time complexity of the FTDSM-GTMoE model is primarily $\mathcal{O}(W \cdot N^2 \cdot L + W \cdot N^2 \cdot d + W \cdot N \cdot d^2)$. This formulation demonstrates that the framework effectively manages the computational overhead of high-dimensional spatiotemporal modeling. By maintaining polynomial complexity relative to the number of regions and window length, the model theoretically guarantees the efficiency and scalability required for processing large-scale clinical rs-fMRI datasets.

5 Computational Efficiency and Cost Comparison

In practical computer-aided diagnosis scenarios, alongside high classification accuracy, the inference efficiency and system resource consumption of a model are critical determinants for its successful clinical translation. We conducted an empirical efficiency comparison between FTDSM-GTMoE and various advanced mainstream models on the AD vs. NC classification task. To ensure fairness, all tests were performed under a unified hardware and software environment (Linux operating system, PyTorch 1.8 framework, and a single NVIDIA RTX 3090 GPU for inference acceleration). Table 1 reports the total parameter count, the total testing time on the test set, and the single-sample inference latency for each model under five-fold cross-validation.

As shown in Table 1, the FTDSM-GTMoE achieves an excellent balance among classification precision, interpretability, and computational efficiency. Regarding spatial complexity, the parameter count of our model (15.06 M) is significantly lower than that of MSTGAC (22.85 M), an advanced dynamic sequence model that similarly employs a sliding window mechanism, yielding a reduction of approximately 34%. Furthermore, it is substantially more compact than FTF-GNN (68.40 M). This parameter compression advantage is primarily attributed to the hardware-aware characteristics of the Mamba architecture and the sparse activation mechanism of the GT-MoE module.

In terms of time cost and inference speed, the single-sample inference latency of FTDSM-GTMoE is approximately 21.23 ms. Although this is objectively higher than

Table 1: Computational efficiency and cost comparison between state-of-the-art methods and FTDSM-GTMoE.

Method	Params / M	Total Time / s	Latency / ms
BrainGNN	0.068	0.0459 ± 0.004	0.528 ± 0.05
HFBN	0.334	0.0695 ± 0.030	0.799 ± 0.35
ACIFBN	1.648	0.2512 ± 0.038	1.443 ± 0.44
BNT	2.964	0.0308 ± 0.002	0.178 ± 0.01
Com-BNTF	10.809	0.1260 ± 0.003	0.724 ± 0.03
BolT	1.473	0.1984 ± 0.015	1.195 ± 0.18
MSTGAC	22.853	1.9289 ± 0.026	22.171 ± 0.30
FTF-GNN	68.400	0.0322 ± 0.008	0.371 ± 0.10
Ours	15.064	1.8467 ± 0.011	21.226 ± 0.12

that of traditional models (e.g., BrainGNN) which only extract macroscopic static graph topologies, this moderate increase in computation is a worthwhile trade-off for accurately capturing the non-stationary evolutionary patterns of functional brain networks and yielding highly valuable spatiotemporal dual-dimensional interpretability. Considering that the clinical acquisition of rs-fMRI data typically takes several minutes, and that the intelligent analysis of neuroimaging is predominantly an offline batch process, an inference speed of 20 milliseconds per sample (processing nearly 50 subjects per second) is virtually instantaneous. Such processing capability is fully competent to meet the high-throughput concurrent diagnostic demands in real-world medical environments.

Data and Code Availability

The raw rs-fMRI data used in this study are publicly available from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database at <https://adni.loni.usc.edu>.

We also provided the processed data and source codes at: <https://github.com/weiba/FTDSM-GTMoE>.

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